

PreOne UI Design Philosophy

PreOne Fluent Metro Design System — Enterprise Preschool Operating System

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Introduction

Document Purpose

This UI Design Philosophy document is the authoritative reference for how PreOne looks, feels, and behaves across every screen, every module, and every user role. It defines the PreOne Fluent Metro Design System — an original synthesis of Windows 8 Metro's tile-based hierarchy, Windows 11 Fluent's depth and motion, Apple's whitespace discipline, and Google Material's usability conventions. The system is not a copy of any one tradition; it is engineered specifically for the Indian preschool context, where users range from tech-comfortable young teachers to first-time app users among parents and grandparents.

This document covers eight parts: design foundations (principles, typography, color, motion, icons, illustrations), the Theme Engine (PreOne's major USP), the PreOne Design Studio (the customization module), the layout system, the 86-component library, the complete page catalog across all 14 modules, multi-tenant branding and white-label architecture, and the implementation roadmap. Three appendices provide the full design token reference, page catalog summary, and glossary.

Design System Name

PreOne Fluent Metro Design System. 'Fluent' for the depth, motion, and focus indicators inherited from Windows 11 Fluent Design. 'Metro' for the bold, tile-based, content-first composition inherited from Windows 8 Metro. The name signals that this is a modern, layered, motion-aware system — not a flat minimalism, not a heavy skeuomorphism. It is the synthesis that works for preschool ERP: warm enough for parents, structured enough for admins, fast enough for teachers in the classroom.

Design Influences

Every design system stands on the shoulders of predecessors. PreOne Fluent Metro is transparent about its influences and explicit about what it borrows, what it rejects, and how it synthesizes them into something original. The table below documents the four primary influences and how each contributes to the final system.

Source	Borrowed	Rejected	Synthesis
Windows 8 Metro	Tile-based information hierarchy, bold typography, content-first composition	Hard edges, limited color palette, cold feel	Tile-based layout with rounded corners, warmer palette, and softer elevation
Windows 11 Fluent	Material depth, acrylic surfaces, smooth motion, focus indicators	Heavy translucency on low-end hardware, complex acrylic rendering cost	Soft elevation via 1-2px shadows, smooth 150-250ms motion, visible focus rings for accessibility
Apple HIG	Whitespace discipline, typographic restraint, progressive disclosure, delight in detail	Platform-locked aesthetics, premium-only positioning	Generous whitespace (max 6 cards per screen), restrained type scale, progressive disclosure for complex forms
Google Material	Elevation tokens, recognizable iconography, standard motion curves, predictable component behavior	Overly busy iconography, dense default spacing, generic feel	Outlined minimal icons, generous spacing, branded motion, PreOne-specific component semantics

Document Hierarchy

Document	Version	Relationship	Authority
Vision Document	v1.0	Predecessor	Product & business direction
Master PRD	v1.0	Predecessor	Product requirements
DDD Document	v1.0	Predecessor	Domain model & bounded contexts
Engineering Standards	v1.0	Predecessor	Engineering execution
UI Design Philosophy	**v1.0**	**This document**	**Visual & interaction design**

Document	Version	Relationship	Authority
Component Library	v1.0 (TBD)	Successor	React/React Native implementation
Design Studio	v1.0 (TBD)	Successor	Theme customization module
Parent App Design	v1.0 (TBD)	Successor	PreOne Hub mobile design

Scope Summary

This document specifies 14 modules, 320 page types, 86 components, and the complete design token architecture. It covers the web portal (Next.js), the parent mobile app (React Native), and the design tokens that keep web, mobile, and Figma in sync. It does not cover marketing websites (separate brand guidelines) or the public preone.in website (separate design). It covers everything a user sees after logging in to PreOne.

Dimension	Count
Modules (Bounded Contexts)	14
Page Types (across all modules)	320
Components (in library)	86
Design Tokens (color, spacing, radius, shadow, type, motion)	75+
Responsive Breakpoints	5 (sm, md, lg, xl, 2xl)
Theme Hierarchy Levels	4 (platform, school, branch, user)
Branding Levels	4 (Standard, Premium, Enterprise, White-Label)
Supported Fonts	4 (Inter, Poppins, Nunito, Roboto)
WCAG Compliance Target	Level AA

Part 1 — Design Foundations

The PreOne Fluent Metro Design System is an original design language built on the shoulders of three industry-defining traditions: the bold tile-based hierarchy of Windows 8 Metro, the depth and motion of Windows 11 Fluent, and the calm whitespace discipline of Apple. It is not a copy of any one system; it is a synthesis deliberately engineered for the Indian preschool context — where users range from tech-comfortable young teachers to first-time app users among parents and grandparents. Every principle, token, and component in this document serves the same goal: making the complex simple, the heavy light, and the impersonal warm.

1.1 Five Design Principles

The PreOne Fluent Metro Design System rests on five principles. They are not equal — they are ordered. 'Fresh' comes first because without whitespace, nothing else matters. 'Tile Based' comes second because without glanceability, dashboards fail. 'Everything in Cards' comes third because without containers, consistency breaks. 'Rounded UI' and 'Flat' come fourth and fifth — they are aesthetic refinements that make the system feel modern, but they build on the first three. Every principle has a rule and an anti-pattern; engineers and designers reference both.

1. Fresh

Principle: Lots of white space. Screen कधीच भरलेला वाटू नये.

Whitespace is not empty space — it is the breathing room that lets content speak. Every screen reserves at least 30% of its area for whitespace. Cards are separated by 24-32px gaps. Section headers have generous top padding (48-64px). A screen that feels 'full' is a screen that needs to be split. This principle fights the Indian SaaS tendency to cram every pixel with information, which overwhelms non-technical users and reduces comprehension.

Rule: Maximum 6 cards + 1 table + 1 chart per screen. More than that? Split the screen or move to a detail page.

Anti-pattern: Dashboard with 12 widgets, 3 tables, and 2 charts on one screen. User cannot tell what matters first.

2. Tile Based

Principle: Windows 8 सारखे पण Modern. Information in glanceable tiles, not just rows.

Tiles are the primary information unit. Each tile communicates one piece of information at a glance — today's attendance (95%), pending fees (₹42,500), new admissions (12). Tiles have a clear hierarchy: large number (the 'what'), small label (the 'context'), and a 'View →' action (the 'next step'). Tiles are sized by importance, not by symmetry — a critical metric gets a larger tile than a secondary one. This is the modern evolution of the Windows 8 Start screen: same information density, but with rounded corners, softer colors, and richer interactions.

Rule: Every dashboard tile answers one question in one glance: number + label + view-link.

Anti-pattern: Tile that tries to show attendance AND fees AND admissions — user doesn't know what to look at first.

3. Everything in Cards

Principle: Forms सुद्धा, Lists सुद्धा, Charts सुद्धा, Reports सुद्धा — everything lives in a card.

Cards are the universal container. A form is a card. A list is a card. A chart is a card. A report is a card. This consistency gives users a single mental model: 'I am looking at a card, and a card has a title, content, and optional actions.' Cards have rounded corners (12-16px), soft elevation (1-2px shadow), and a header-content-action structure. The card header carries the title and optional filter/sort controls; the body holds the content; the footer (optional) holds primary actions. This pattern repeats across all 320 page types in PreOne.

Rule: If content needs a container, it goes in a card. Cards never nest more than 2 levels deep.

Anti-pattern: Form fields scattered on a page without visual grouping; chart floating without a title or context.

4. Rounded UI

Principle: Not sharp. Radius 12-16px on cards, 8-12px on inputs, 50% on avatars, 24px on large containers.

Rounded corners feel friendly and modern — critical for a preschool product where warmth matters. Sharp corners (radius 0) feel corporate and cold; we avoid them except for data-dense tables where edge alignment matters. The default card radius is 16px, inputs use 12px, buttons use 8-10px depending on size, and avatars are fully circular. The radius scale is a design token — admins can adjust it via Design Studio (8px / 12px / 16px presets), and the entire UI updates instantly.

Rule: Default radius: cards 16px, inputs 12px, buttons 8px, avatars 50%. Design Studio allows global radius adjustment.

Anti-pattern: Inconsistent radii across the same screen — card at 16px, button at 4px, input at 0px. Looks broken.

5. Flat

Principle: No heavy shadows. Very soft elevation. Depth via subtle layering, not drop shadows.

Heavy shadows (4px+ blur, 30%+ opacity) feel dated and Web 2.0. PreOne uses soft elevation: 1-2px shadows with 6-12px blur and 4-8% opacity. Depth is communicated through subtle background tint (surface tokens) and 1px borders, not through shadow drama. The only exception is modals and popovers, which use slightly stronger elevation (3-4px, 12% opacity) to communicate layering above the page. This keeps the UI feeling modern, calm, and lightweight — essential for low-end devices common in Indian schools.

Rule: Card shadow: 0 1px 3px rgba(0,0,0,0.04). Modal shadow: 0 4px 12px rgba(0,0,0,0.12). No other shadows.

Anti-pattern: Card with 0 8px 24px shadow — feels heavy, slows rendering on low-end devices, looks Web 2.0.

1.2 Typography

Inter is the default because it is optimized for screen reading at small sizes, has excellent multilingual coverage (Latin, Devanagari via fallback), and renders crisply across Windows, Mac, Android, and iOS. Poppins is offered as an alternative for schools wanting a rounder, friendlier feel; Nunito for soft warmth; Roboto for familiarity (Android-native). The admin selects the font family in Design Studio; CSS variables propagate the choice across the entire UI.

Default font: Inter. Alternatives: Poppins, Nunito, Roboto. 1.5 for body text, 1.2 for headings, 1.4 for table cells. Generous line height improves readability for non-native English readers.

Type Scale

Token	Size	Weight	Usage
--text-xs	12px	400	Captions, timestamps, secondary metadata
--text-sm	14px	400	Table cells, form helper text, list items
--text-base	16px	400	Body text, default input text, paragraph
--text-lg	18px	500	Card titles, section subheadings, form labels
--text-xl	20px	600	Page subheadings, dialog titles, important tiles
--text-2xl	24px	600	Page titles, primary tile numbers
--text-3xl	32px	700	Dashboard tile numbers, hero text
--text-4xl	40px	700	Login screen school name, onboarding hero

Font Weights

Weight	Name	Usage
400	Regular	Body text, table content, helper text
500	Medium	Card titles, form labels, button text
600	Semibold	Page titles, section headers, important tiles
700	Bold	Hero numbers, login branding, onboarding headlines

1.3 Color Engine

Instead of hardcoding #6A35FF everywhere, every color is a CSS variable (design token). When the theme changes, the complete UI updates instantly. This is the technical foundation of multi-tenant branding: one codebase, infinite themes.

Override mechanism: School theme JSON is loaded at app boot and injected as inline CSS variables on :root. Branch themes inject on a more specific selector. User preference (light/dark) applies a class on <html>. The cascade resolves: user preference > branch > school > platform. No component code reads hardcoded colors — all components reference var(--token).

Semantic Color Tokens (Light Mode)

Token	Default	Usage
--primary	#6A35FF	Primary actions, active states, brand emphasis, links
--primary-hover	#5A2BD9	Hover state for primary buttons and links
--primary-light	#F0EBFF	Subtle backgrounds, selected item tints, chip backgrounds
--secondary	#009688	Secondary actions, supporting elements, badges
--accent	#FF6B35	Highlights, call-to-action accents, notification dots
--success	#10B981	Success states, completed items, positive metrics
--warning	#F59E0B	Warnings, pending items, attention-needed states
--danger	#EF4444	Errors, destructive actions, overdue items
--info	#3B82F6	Informational badges, informational toasts, tips
--bg-page	#F8F9FB	Page background behind cards
--bg-card	#FFFFFF	Card background, input background on light surfaces

Token	Default	Usage
--bg-surface	#F1F3F5	Secondary surfaces, table header rows, disabled states
--text-primary	#1A1D21	Primary text, headings, important content
--text-secondary	#5B6B7D	Secondary text, captions, helper text
--text-muted	#9AA5B1	Muted text, placeholders, disabled labels
--border-default	#E3E7EB	Default borders on cards, inputs, dividers
--border-strong	#C4CAD2	Stronger borders for emphasis, table outer borders

Dark Mode Token Overrides

Token	Dark Value	Usage
--bg-page	#0F1419	Dark mode page background
--bg-card	#1A1F26	Dark mode card background
--bg-surface	#252B33	Dark mode secondary surface
--text-primary	#F1F3F5	Dark mode primary text
--text-secondary	#9AA5B1	Dark mode secondary text
--border-default	#2D3440	Dark mode default border

1.4 Motion

150-250ms for all transitions. Nothing flashy, nothing slow. Users should feel the UI respond instantly, not wait for animations to finish. cubic-bezier(0.4, 0, 0.2, 1) for most transitions (Material standard). cubic-bezier(0.0, 0, 0.2, 1) for entrances (deceleration). cubic-bezier(0.4, 0, 1, 1) for exits (acceleration).

Motion Rules

- Hover transitions: 150ms ease-out. Buttons, links, cards lift subtly on hover.

- Modal/Dialog open: 200ms with fade + 8px slide-up. Close: 150ms fade + 4px slide-down.
- Tile hover: 200ms lift (transform: translateY(-2px)) + shadow enhance. Communicates clickability.
- Page transitions: 200ms fade. No slide — slides disorient in dense admin UIs.
- Loading skeletons: 1.5s shimmer animation. Pulse, not spin — feels calmer.
- Toast notifications: 250ms slide-in from top-right, 4s display, 200ms slide-out.
- Reduced motion (accessibility): all transitions collapse to 0ms. Respects user's OS setting.

Forbidden Motion

- Spinning loaders (use skeleton screens instead — they communicate shape, not just 'wait')
- Bounce animations (feel childish for an enterprise product)
- Animations longer than 300ms (feel sluggish)
- Multiple simultaneous animations on different elements (chaotic)
- Parallax scrolling (performance cost, disorienting)
- Auto-playing carousels (users miss content, accessibility issue)

1.5 Iconography

Style: Outlined, minimal, rounded. No colorful icons. No filled icons except for active/selected states.

Library: Lucide (open-source, consistent, 1500+ icons). Custom icons only when Lucide lacks a suitable option — must follow the same stroke width (1.5px), corner radius (2px), and 24x24 viewBox.

Sizing: 16px (inline in text), 20px (buttons, list items), 24px (navigation, cards), 32px (empty states), 48px (onboarding, large empty states).

Icon Rules

- Stroke width: 1.5px default, 2px for active/selected state
- Color: inherit from parent text color. Never hardcode icon color.
- Active state: filled variant or primary color tint
- No emoji in production UI (inconsistent across platforms, accessibility issues)

- No decorative icons without text labels (accessibility: icons need labels for screen readers)

Forbidden Icons

- Material filled icons (too heavy, inconsistent with outlined style)
- Font Awesome (overused, less consistent than Lucide)
- Custom hand-drawn icons (inconsistent stroke width, alignment issues)
- Colorful multi-tone icons (breaks the minimal aesthetic)

1.6 Illustrations

Style: Soft, friendly, kids-themed but professional. Hand-drawn feel with consistent stroke. Warm colors from the brand palette. Used in empty states, onboarding, error states, and welcome screens — never in dense data views.

Library: Custom illustrations commissioned from a professional illustrator (unDraw style but warmer). Stored as SVG components in the design system package. 40 base illustrations covering: empty states (no students, no fees, no messages), error states (network error, permission denied, not found), onboarding (welcome, first steps, completion), and seasonal (festive greetings for Diwali, Christmas, etc.).

Illustration Usage

- Empty states: illustration + headline + subtext + primary action button
- Error states: illustration + headline + retry button
- Onboarding: full-screen illustration + step indicator + next button
- Welcome screen: school-branded illustration + welcome message + get-started button
- Never: in tables, forms, dashboards, or anywhere they compete with data

Part 2 — Theme Engine

The Theme Engine is PreOne's major unique selling proposition. In one database, 1000 schools can each have a completely different branded UI — different logos, colors, fonts, dashboard layouts — while sharing the same backend codebase and database schema. This is multi-tenant branding at the presentation layer, achieved through a four-level theme hierarchy and CSS variable (design token) propagation. No component code changes; only theme tokens change. This feature alone positions PreOne apart from every other preschool ERP in the Indian market, where 'customization' usually means a different logo on a login page.

2.1 Four-Level Theme Hierarchy

Platform Theme ↓ Subscription Plan (limits which features are themeable) ↓ School Theme ↓ Branch Theme ↓ User Preference. The cascade resolves at runtime: when a user loads the app, the platform theme loads first, then the school theme overrides matching tokens, then the branch theme overrides further (if present), then the user preference applies (light/dark, layout). The final computed theme is injected as CSS variables on :root.

Level	Name	Source	Scope	Configurable
1	Platform Theme	PreOne default	All tenants without a custom theme	No
2	School Branding	School Admin via Design Studio	All branches and users under one school	Yes
3	Branch Branding	Branch Admin via Design Studio (limited)	All users under one branch	Yes
4	User Preference	Each user via Profile → Preferences	Single user only	Yes

Hierarchy Details

Platform Theme (Level 1): The PreOne Fluent Metro default. Purple primary (#6A35FF), teal secondary (#009688), Inter font, 16px radius. This is what a brand-new tenant sees before they customize anything. It is maintained by the PreOne design team and updated with each platform release.

School Branding (Level 2): School logo, school name, school colors (primary, secondary, accent, and semantic colors), school banner, school font. Set by the School Admin in Settings → Branding → Theme. Changes propagate to all branches and all users under that school within 60 seconds (cache invalidation). This is the primary customization layer.

Branch Branding (Level 3): Optional. Branch banner, branch-specific welcome message, limited color overrides (only accent color — primary/secondary remain school-level for brand consistency). Useful for school chains where each branch has a distinct identity (e.g., 'Little Angels — Koramangala' vs 'Little Angels — Indiranagar').

User Preference (Level 4): Light/Dark/Auto mode, sidebar collapsed/expanded, dashboard widget arrangement, font size preference (accessibility). These do not override brand colors — they layer on top. A user in dark mode still sees the school's brand colors, just rendered against a dark background.

2.2 Theme Configuration

Path: Settings → Branding → Theme

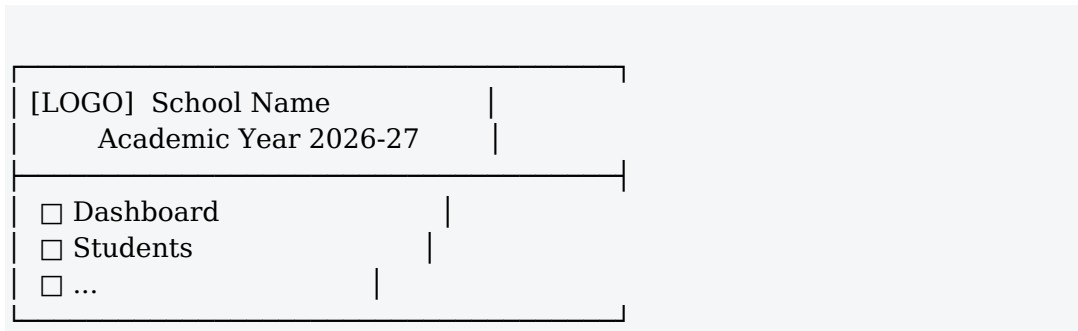
Access: School Admin role or higher. Branch Admin can configure branch-level overrides only.

Configuration Operations

- Change Logo — Upload PNG or SVG. Auto-generates favicon. Stored in S3, CDN-cached.
- Change Colors — Primary, Secondary, Accent, Success, Warning, Danger, Info. Live preview shows old vs new.
- Change Font — Inter (default), Poppins, Nunito, Roboto. Applied globally.
- Change Border Radius — 8px (compact), 12px (default), 16px (comfortable). Global token update.
- Change Layout — Sidebar collapsed/expanded default, dashboard widget arrangement.
- Live Preview — Old theme → New theme comparison before Apply.
- Apply — Commits changes. Cache invalidated. All users see new theme within 60 seconds.
- Revert — One-click revert to previous theme. Safety net for bad changes.

2.3 Logo Position

Default: Top-left of sidebar (expanded) or center of sidebar (collapsed). School name to the right of logo. Academic year below school name in smaller, muted text.



Logo Rules

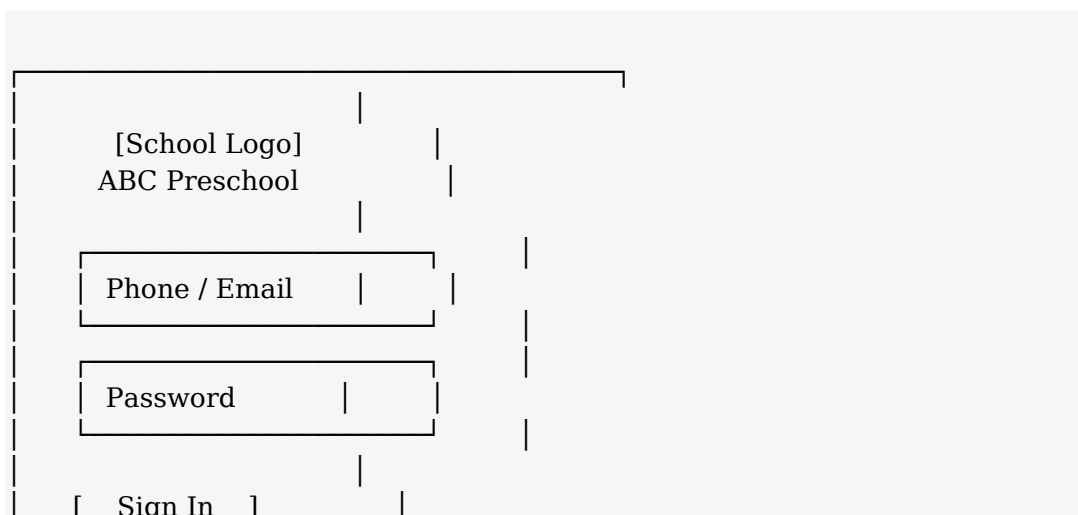
- Logo max-height: 40px (expanded sidebar), 32px (collapsed sidebar)
- Logo aspect ratio preserved — no stretching
- School name truncated with ellipsis if too long (tooltip shows full name)
- Academic year shown only when user has multi-year access
- On mobile: logo top-center, school name below, no academic year

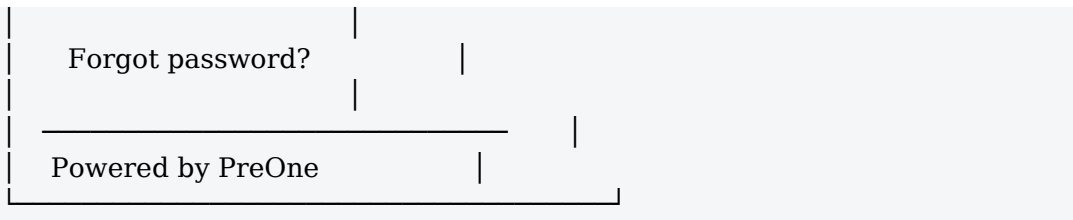
2.4 Login Screen

Default: PreOne logo on login screen for tenants without custom branding.

Branded: School logo on login screen when school theme is configured. 'Powered by PreOne' badge at bottom — required on all plans except Enterprise White-Label.

Login Screen Example





Login Screen Rules

- School logo centered at top (max 80px height)
- School name below logo in --text-2xl, --text-primary
- Login form card below, max-width 400px, centered
- 'Powered by PreOne' badge at bottom: PreOne logo + text, 12px, --text-muted
- Enterprise White-Label plan: 'Powered by PreOne' badge removable
- Background: subtle pattern or school-uploaded login background image (optional, Design Studio feature)

2.5 Theme Storage

Location: PostgreSQL: tenants.theme_config (JSONB), branches.theme_config (JSONB), user_preferences.theme (JSONB)

Theme JSON Schema

```
{
  "theme": "light" | "dark" | "auto",
  "colors": {
    "primary": "#009688",
    "secondary": "#FF6B35",
    "accent": "#FFD23F",
    "success": "#10B981",
    "warning": "#F59E0B",
    "danger": "#EF4444",
    "info": "#3B82F6"
  },
  "logo": {
    "url": "https://cdn.preone.in/schools/abc/logo.png",
    "favicon": "https://cdn.preone.in/schools/abc/favicon.ico"
  },
  "font": "Poppins",
  "border_radius": 12,
  "layout": {
    "sidebar": "expanded" | "collapsed" | "auto-hide",
    "dashboard": ["attendance", "fees", "admissions", "reports"]
  },
  "login_background": "https://cdn.preone.in/schools/abc/login-bg.jpg",
}
```

```
"welcome_message": "Welcome to ABC Preschool!"  
}
```

Caching: Theme JSON cached in Redis (key: theme:{tenantId}:{branchId}). TTL 5 minutes.
Cache invalidated on theme update via event. CDN serves logo/favicon with 1-year cache + content-hash URL.

Part 3 — PreOne Design Studio

PreOne Design Studio is a dedicated module that elevates PreOne to a premium position in the Indian preschool ERP market. Accessible at Settings → Design Studio, it allows a School Admin to customize the entire visual identity of their PreOne instance — logo, colors, fonts, layouts, border radius, dark mode, welcome messages, banners, favicons, and even limited parent-app branding — without writing a single line of code. The studio operates entirely on CSS variables (design tokens), so no component code changes; only theme tokens change, and the entire web portal and PreOne Hub parent app re-render in the new branding instantly. This is the feature that lets 1000 schools share one codebase while each having a unique branded experience — a positioning no competitor in the Indian market currently offers.

3.1 Access Matrix

Design Studio is accessible to School Admin and above, with Branch Admin having limited scope. The access matrix below defines who can do what.

Role	Access	Notes
Super Admin	All features + platform default theme	PreOne internal team only
School Admin	All Design Studio features for their school	Primary user of Design Studio
Branch Admin	Branch banner + welcome message + accent color only	Limited to branch-scope overrides
Teacher / Staff	View-only (cannot modify)	Cannot access Design Studio
Parent	No access	Parents see the configured theme, cannot change it

3.2 Design Studio Features

Design Studio provides 13 customization features. Each feature operates on CSS variables (design tokens) — no component code changes. The school admin configures in the studio, previews live, and applies. All users under the school see the new theme within 60 seconds.

School Logo

Upload PNG (max 1MB, min 200x200px) or SVG (max 500KB). Auto-generates favicon at 32x32, 64x64, 180x180 (Apple touch icon). Stored in S3, served via CDN with content-hash URL for cache-busting. Live preview shows logo in sidebar, login screen, header, and parent app before apply.

Constraints: PNG with transparency preferred. SVG preferred for crisp rendering at all sizes. School name text shown alongside logo — logo should not contain the school name to avoid redundancy.

Login Background

Upload JPG/PNG (max 2MB, recommended 1920x1080). Applied as login screen background with 30% dark overlay for text readability. Default: subtle gradient using brand colors. Live preview shows login screen with the uploaded background.

Constraints: No text in the background image (overlay will distort). No student photos (privacy + consent). Recommended: abstract patterns, school building exterior, or classroom illustrations.

Primary / Secondary Colors

Color picker for primary, secondary, accent, success, warning, danger, info. Each color validated for WCAG AA contrast against white background (minimum 4.5:1 for text, 3:1 for large text and UI components). Invalid colors show a warning with suggested alternatives. Live preview shows the entire UI in the new colors before apply.

Constraints: Primary color must have a hover variant (auto-generated by darkening 10%). Primary light variant auto-generated at 10% opacity on white. Cannot use pure black or pure white as primary.

Dashboard Layout

Drag-and-drop tile arrangement. Admin chooses which tiles appear on each role's dashboard (Principal, Teacher, Parent, Accountant, etc.). Tile sizes adjustable (small/medium/large). Tiles can be hidden, pinned, or marked as favorite (always top). Default layouts provided per role; admin customizes from there.

Constraints: Maximum 12 tiles per dashboard (beyond that, split into tabs). Critical tiles (e.g., 'Today's Attendance' for Principal) cannot be hidden — only repositioned. Layout saved per role, not per user.

Sidebar Style

Three options: Expanded (default, 240px wide, icon + label), Collapsed (72px wide, icon only with tooltip), Auto-Hide (overlays on hover, full content width otherwise). Per-user preference overrides admin default. Sidebar items customizable per role (e.g., parents see only relevant items).

Constraints: Auto-Hide disabled on mobile (always overlay on mobile). Minimum 6 items visible in sidebar — if fewer, consider consolidating. Logout and Settings always at bottom.

Card Style

Two density options: Comfortable (default, generous padding, 16px radius) and Compact (tighter padding, 12px radius, more content per screen). Comfortable is preferred for touch devices and less-technical users; Compact is preferred by power users on desktops. Per-user preference overrides admin default.

Constraints: Compact mode reduces card padding from 24px to 16px, table row height from 48px to 36px. Font sizes unchanged. Critical for users on small monitors.

Border Radius

Three presets: 8px (Sharp — more corporate), 12px (Default — balanced), 16px (Soft — friendliest). Applied globally to cards, inputs, buttons, modals. Live preview shows the entire UI in the new radius before apply.

Constraints: Avatars always 50% (circular) regardless of global radius. Modals always use global radius + 4px (slightly softer than cards).

Font Family

Four options: Inter (default), Poppins (rounder, friendly), Nunito (soft, warm), Roboto (familiar, Android-native). All four pre-loaded as web fonts (Google Fonts) for instant switching. Applied globally to all text. Devanagari (Marathi/Hindi) support verified for all four via Noto Sans Devanagari fallback.

Constraints: Custom font upload not supported in v1 (planned for v2). All four fonts have weights 400, 500, 600, 700 available. Font loading optimized via font-display: swap.

Light / Dark / Auto Mode

Three options: Light (default), Dark (inverted background/text, brand colors stay), Auto (follows OS setting). Per-user preference; admin sets the default. Dark mode uses the dark token values defined in Part 1 — brand colors remain, only surfaces and text invert.

Constraints: Dark mode contrast verified for WCAG AA. Charts and illustrations have dark-mode variants. Printed reports always light mode (dark mode would waste ink).

Welcome Message

Custom welcome message shown on login screen below school name (max 80 chars). Default: 'Welcome to {School Name}'. Supports English + Devanagari. Used to set the tone — e.g., 'Where little ones bloom' or 'नन्हें कदम, बड़े सपने'.

Constraints: No emojis in welcome message (inconsistent across platforms). No HTML (plain text only). Profanity filter applied on save.

School Banner

Upload JPG/PNG (max 3MB, recommended 1920x400). Shown on login screen (above login form) and on the main dashboard header for the first 3 seconds after login (then collapses). Default: gradient using brand colors. Used for seasonal greetings (Diwali, Christmas, New Year) — admin can schedule banner changes.

Constraints: Banner should not contain critical text (collapses on small screens). Mobile: banner hidden, only welcome message shown. Banner scheduling: admin sets start/end date for each banner.

Favicon

Auto-generated from school logo. Manual override available (upload .ico or .png). Applied to browser tab, bookmark, PWA home screen icon. Multiple sizes generated automatically (16x16, 32x32, 180x180, 192x192, 512x512).

Constraints: Auto-generation prefers the logo's most distinctive element (may require admin to upload a simplified favicon separately if the full logo is too detailed).

Parent App Branding (Limited)

School logo and school name shown in PreOne Hub parent app (mobile). Parent app uses school colors for primary actions. Login screen of parent app shows school logo + 'Powered

by PreOne'. Limited because the parent app is a shared binary — full white-label parent app requires Enterprise plan.

Constraints: Enterprise plan: fully white-labeled parent app (custom app name, custom app icon, custom splash screen, no 'PreOne' mention). Available at additional cost.

3.3 Design Studio Workflow

The workflow below describes the end-to-end flow from opening Design Studio to all users seeing the new theme. The key insight is the live preview — admin sees exactly what users will see before committing. Revert provides a safety net for bad changes.

- 1. Admin navigates to Settings → Design Studio
- 2. Admin sees current theme on the left, live preview on the right
- 3. Admin modifies any combination of logo, colors, fonts, layout, radius, etc.
- 4. Live preview updates instantly as admin makes changes (no save required to preview)
- 5. Admin clicks 'Compare' to see old vs new side-by-side
- 6. Admin clicks 'Apply' to commit changes — confirmation dialog warns that all users will see the new theme
- 7. On confirm: theme JSON updated in DB, Redis cache invalidated, CDN cache busted (logo URL changes)
- 8. Within 60 seconds, all active users see the new theme on their next API call or page navigation
- 9. Admin can 'Revert' to previous theme at any time (one-click, no confirmation — revert is always safe)

3.4 Technical Foundation

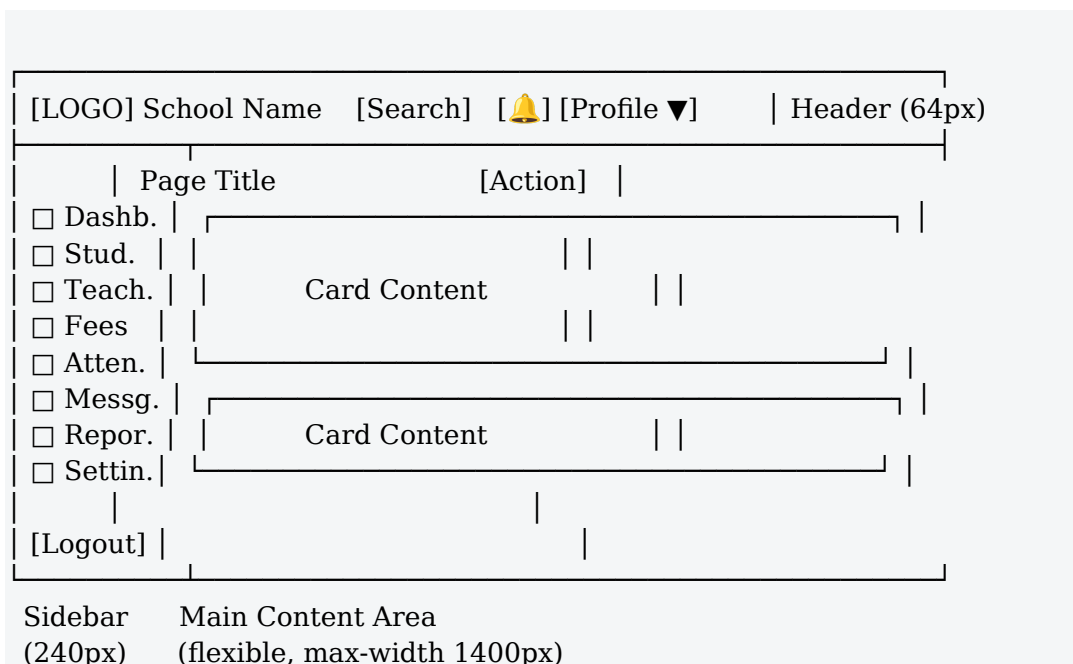
Design Studio operates entirely on CSS variables (design tokens). The theme JSON stored in DB is transformed into CSS variables and injected on :root at app boot. When a user navigates, the theme is included in the initial HTML response (server-side rendering) — no flash of unstyled content. Components reference var(--token), never hardcoded values. This is why a theme change updates the entire UI without any component code changes — the components don't know about themes; they just read CSS variables.

Part 4 — Layout System

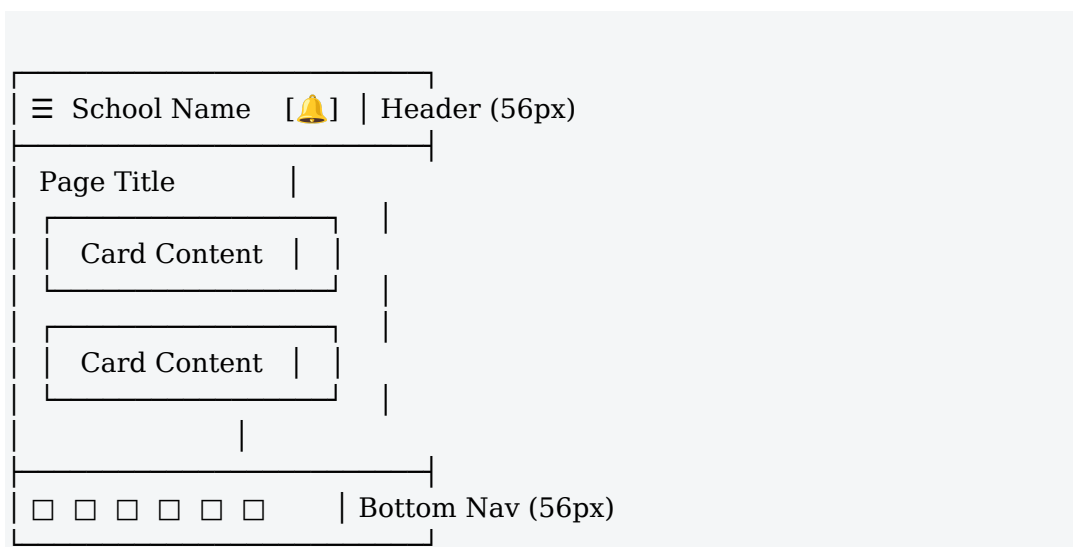
The layout system defines how pages, sidebars, headers, and content areas are structured across PreOne. Consistency in layout reduces cognitive load — once a user learns the layout, they can navigate any page in the system. The layout system is responsive (desktop, tablet, mobile), adaptive (sidebar behavior changes by screen size), and role-aware (different roles see different navigation items).

4.1 Global Layout

Desktop Layout



Mobile Layout



4.2 Header

Height: 64px desktop, 56px mobile

Header Contents

- Logo + School Name (left, sidebar collapsed — full school name; sidebar expanded — just logo)
- Global Search (center, 400px max-width, searches students, staff, invoices by name/ID/phone)
- Notification bell (right, shows unread count badge, dropdown shows recent notifications)
- Profile menu (right, avatar + name + role, dropdown: Profile, Settings, Theme, Logout)

Header Rules

- Header is sticky (position: sticky, top: 0) — always visible on scroll
- Header background: var(--bg-card) with 1px bottom border var(--border-default)
- Header z-index: 100 (above content, below modals at 1000)
- On mobile: logo hidden, only school name; search collapses to icon; bottom nav replaces sidebar

4.3 Sidebar

Width: 240px expanded, 72px collapsed, 0px auto-hide (overlay on hover)

Sidebar Contents

- Logo + School Name + Academic Year (top, 64px tall, matches header)
- Navigation items (middle, grouped by section: Operations, Academics, Administration, Reports, Settings)
- Help + Logout (bottom, 48px tall)

Sidebar Behaviors

- Expanded (default on desktop ≥ 1200 px): full sidebar visible, icon + label
- Collapsed (default on tablet 768-1199px or user preference): icon only, expand on hover (overlay)

- Auto-Hide (user preference): sidebar hidden, hamburger menu reveals overlay sidebar
- Mobile (< 768px): no sidebar, bottom navigation bar with 5 primary items + 'More'

Sidebar Customization

- Admin can reorder navigation items per role (Design Studio)
- Admin can hide non-critical items per role (e.g., hide 'Inventory' for teachers)
- Critical items (Dashboard, Logout, Settings) cannot be hidden
- User can collapse/expand sidebar (preference saved per user)

4.4 Content Area

Max width: 1400px (centered on screens wider than 1400px, with 32px page padding)

Padding: 32px desktop, 24px tablet, 16px mobile

Structure: Page Title (top, with optional action button on right) → Breadcrumb (optional, below title) → Filter Bar (optional, below breadcrumb) → Cards (stacked vertically, 24px gap) → Pagination (optional, at bottom of lists)

White Space Rule

Rule: Maximum 6 cards + 1 table + 1 chart per screen

Detail: More than that? Split the screen into tabs or move secondary content to a detail page. The goal is glanceability — a user should be able to understand the page's purpose in 3 seconds.

Enforcement: Design review rejects mockups with more than 6 cards. Engineering review rejects PRs with more than 6 cards on a single route without tab splitting.

4.5 Responsive Breakpoints

Breakpoint	Min Width	Target	Layout
sm	0	Mobile phones (portrait)	Single column, bottom nav, stacked cards
md	768	Tablets (portrait), Mobile phones (landscape)	Single column, collapsible sidebar, stacked cards

Breakpoint	Min Width	Target	Layout
lg	1024	Tablets (landscape), Small laptops	Two-column where appropriate, expanded sidebar
xl	1280	Standard laptops, Small desktops	Multi-column grids, expanded sidebar
2xl	1536	Large desktops, Wide monitors	Multi-column grids, expanded sidebar, max-width 1400px content

4.6 Grid System

Type: 12-column CSS Grid with 24px gutter

Usage: Cards align to grid columns. A card can span 12 (full width), 6 (half), 4 (third), 3 (quarter), 8 (two-thirds), or 9 (three-quarters). On mobile, all cards span 12 (full width, stacked).

Grid Examples

- Dashboard: 4 stat tiles (3 cols each) + 1 chart (8 cols) + 1 list (4 cols)
- Student list: 1 table (12 cols)
- Student detail: 1 photo + summary (4 cols) + 1 tabs card (8 cols)
- Settings: 1 nav (3 cols) + 1 form (9 cols)

Part 5 — Component Library

The PreOne component library is the single source of truth for UI building blocks. Every component is built once, tested once, and reused across all 320 page types. Components are implemented in React (web), React Native (mobile), and Figma (design) — all three stay in sync via the design tokens. Engineers do not build custom components when a library component exists; they use the library. New components require design review and architecture approval.

5.1 Component Categories

The PreOne component library has 68 components across 8 categories. Each component is built once, tested once, and reused across all 320 page types. Engineers do not build custom components when a library component exists.

Category	Component Count
Layout	12
Form	14
Data	13
Chart	7
Feedback	7
Action	6
Navigation	4
Utility	6

5.2 Layout Components

Component	Key Props	Usage
Card	title, subtitle, actions, padding, radius, elevation	Universal container for all content. Header (title + actions) + Body (content) + optional Footer (actions).

Component	Key Props	Usage
Tile	label, value, icon, trend, onClick, size	Dashboard glanceable metric. Large value (number), small label, optional trend indicator, optional icon. Clickable to drill down.
PageHeader	title, subtitle, breadcrumb, actions	Top of every page. Title + optional subtitle + optional breadcrumb + optional action button on right.
Sidebar	items, collapsed, onItemClick	Left navigation. Items grouped by section. Collapsible to icon-only. Auto-hide on mobile.
Header	logo, schoolName, search, notifications, profile	Top bar. Sticky. Contains brand, search, notifications, profile menu.
BottomNav	items, active	Mobile navigation. 5 primary items + 'More'. Fixed to bottom.
Tabs	tabs, activeTab, onChange	Content organization within a page. Horizontal tabs at top of card. Used when a page has 3-7 sections.
Accordion	items, allowMultiple	Collapsible sections. Used for long forms, FAQs, settings groups.
Modal	open, title, children, actions, size	Overlay dialog for focused tasks (confirm, edit single item). Sizes: sm (400px), md (600px), lg (800px), xl (1000px).
Drawer	open, side, title, children, width	Side panel for filters, details, quick-edit. Slides in from right (default) or left.
Grid	columns, gap, children	12-column responsive grid. Cards align to columns.

5.3 Form Components

Component	Key Props	Usage
TextInput	label, value, onChange, error, helperText, placeholder, required, disabled	Standard text input. Label above, input below, helper text or error below that.
NumberInput	label, value, onChange, min, max, step, prefix, suffix	Numeric input with optional prefix (₹) or suffix (kg, cm). Validates numeric.
Select	label, value, options, onChange, searchable, multi	Dropdown select. Single or multi. Searchable for lists > 10 items.
DatePicker	label, value, onChange, min, max, range	Date selection. Calendar popover. Supports single date or date range.
TimePicker	label, value, onChange, format	Time selection. 12-hour or 24-hour format based on locale.
Checkbox	label, checked, onChange, indeterminate, disabled	Binary selection. Single or in group.
RadioGroup	label, value, options, onChange, direction	Mutually exclusive selection. Horizontal or vertical layout.
Switch	label, checked, onChange, disabled	Toggle for binary settings (on/off). Used in settings, not in forms.
Textarea	label, value, onChange, rows, maxLength, counter	Multi-line text input. Optional character counter.
FileUpload	label, accept, maxSize, multiple, onUpload	Drag-and-drop file upload. Shows preview for images, file name for others. Validates type and size.
RichTextEditor	label, value, onChange, toolbar	WYSIWYG editor for messages, observations, announcements. Limited toolbar (bold, italic, list, link).

Component	Key Props	Usage
OTPIInput	length, value, onChange, onComplete	6-digit OTP input. Individual boxes, auto-advance, paste support.
SearchInput	value, onChange, placeholder, onSearch	Debounced search input with search icon. Triggers search after 300ms of no typing.
FormLayout	columns, children, spacing	Wraps form fields in a responsive grid. 1 column on mobile, 2 on desktop.

5.4 Data Display Components

Component	Key Props	Usage
Table	columns, data, sortable, pagination, selectable, onRowClick	Tabular data. Sticky header. Sortable columns. Pagination at bottom. Row click for detail.
DataTable	columns, data, filters, export, density	Advanced table with filter bar, column show/hide, CSV/PDF export, density toggle. Used for lists.
List	items, renderItem, keyExtractor, loadMore	Vertical list of items. Each item is a row with avatar, title, subtitle, action.
Pagination	page, pageSize, total, onChange	Page navigation. Shows current page, total pages, prev/next, page size selector.
Badge	label, color, variant	Small status indicator. Variants: solid, soft, outline. Colors: success, warning, danger, info, neutral.
Avatar	src, name, size, shape	User photo or initials. Circular (default) or square. Sizes: 24, 32, 40, 48, 64, 96px.

Component	Key Props	Usage
Chip	label, onDelete, onClick, selected	Filter chip, tag, or selection. Removable (x icon) or clickable.
Stat	label, value, unit, trend, trendValue	Inline statistic. Label + value + optional unit + optional trend (up/down arrow with %).
ProgressBar	value, max, color, label	Horizontal progress indicator. Used for form completion, fee collection, attendance.
Timeline	items, variant	Vertical timeline of events. Used for audit logs, application status, admission pipeline.
EmptyState	illustration, title, description, action	Shown when no data. Illustration + headline + subtext + primary action button.
ErrorState	illustration, title, description, onRetry	Shown on error. Illustration + 'Something went wrong' + Retry button.
Skeleton	variant, width, height, count	Loading placeholder. Variants: text, card, table-row, avatar. Shimmer animation.

5.5 Chart Components

Component	Key Props	Usage
LineChart	data, xKey, yKeys, colors, legend	Trend over time. Attendance %, fee collection, admissions trend.
BarChart	data, xKey, yKeys, colors, stacked	Comparison across categories. Branch-wise enrollment, class-wise attendance.

Component	Key Props	Usage
PieChart	data, colors, legend, donut	Composition. Fee breakup, attendance status distribution.
AreaChart	data, xKey, yKeys, colors, stacked	Cumulative trend. Cumulative fees, cumulative admissions.
DonutChart	data, colors, centerLabel	Pie chart with center label. Used for percentage display.
Sparkline	data, color, width, height	Inline mini-chart in tiles and tables. 7-day trend.
GaugeChart	value, max, color, label	Single-value gauge. Attendance rate, target achievement.

5.6 Feedback Components

Component	Key Props	Usage
Toast	message, type, duration, action	Transient notification. Top-right. Types: success, error, warning, info. Auto-dismiss after 4s.
Alert	type, title, message, action	Inline alert in page content. Persistent until dismissed. Types: success, warning, error, info.
Snackbar	message, action, onAction	Bottom-center notification with action button. 'Undo' pattern.
Tooltip	content, position, trigger	Hover/focus tooltip for icon buttons and truncated text.
Popover	content, trigger, position	Rich tooltip with interactive content. Used for quick details without navigation.
ConfirmDialog	title, message, confirmLabel, cancelLabel, onConfirm	Confirmation modal for destructive actions. 'Are you sure?' pattern.

Component	Key Props	Usage
LoadingOverlay	active, message	Full-card loading overlay for long operations (file upload, report generation).

5.7 Action Components

Component	Key Props	Usage
Button	label, variant, size, icon, onClick, loading, disabled	Primary action button. Variants: primary, secondary, outline, ghost, danger. Sizes: sm, md, lg.
IconButton	icon, label, variant, size, onClick	Icon-only button. Tooltip on hover shows label.
FloatingActionButton	icon, label, onClick	FAB for primary create action. Bottom-right. Mobile-only.
ButtonGroup	buttons, value, onChange	Group of toggle buttons. Used for view switch (list/grid), filter selection.
DropdownMenu	items, trigger	Menu of actions. Triggered by button (kebab menu, 'More actions').
Breadcrumb	items, onItemClick	Navigation trail. Home / Students / Raj Patel / Edit.

5.8 Navigation Components

Component	Key Props	Usage
NavLink	to, label, icon, active	Sidebar navigation item. Active state highlighted.
Tabs	tabs, active, onChange	Tab navigation. (Same as Layout Tabs.)
Stepper	steps, current	Multi-step flow indicator. Admission application, onboarding, payment.

Component	Key Props	Usage
Pagination	page, total, onChange	Page navigation. (Same as Data Pagination.)

5.9 Utility Components

Component	Key Props	Usage
Modal	open, onClose, title, children	Overlay dialog. (Same as Layout Modal.)
Drawer	open, onClose, side, children	Side panel. (Same as Layout Drawer.)
Divider	variant, label	Horizontal divider with optional label. Separates sections in cards.
Spacer	size	Vertical spacing element. Use sparingly — prefer card body padding.
Tag	label, color, size	Small label tag. Similar to Badge but smaller, inline.
KeyVal	label, value, copyable	Key-value pair display. Used in detail views. Optional copy-to-clipboard.

Part 6 — Page Catalog (All 14 Modules)

This part catalogs every page type across all 14 PreOne modules. Each module is a bounded context (per DDD v1.0) with its own navigation, dashboards, list views, detail views, and forms. The page catalog serves as the master reference for engineering — every page here must be implemented, tested, and documented. Designers reference this catalog for consistency; product managers reference it for scope; QA references it for test coverage.

6.1 Module Summary

PreOne has 14 modules (bounded contexts), each with its own navigation, dashboards, list views, detail views, and forms. Total page types across all modules: 200. The table below summarizes each module; subsequent sections detail every page in every module.

ID	Module Name	Domain	Pages
M01	Identity & Authentication	Platform	15
M02	CRM (Lead Management)	Admissions	12
M03	Admissions	Admissions	12
M04	Student Lifecycle	Student	13
M05	Academics	Academics	13
M06	Daily Operations (Attendance + Transport)	Attendance + Transport	13
M07	Parent Communication	Communication	12
M08	Finance	Finance	18
M09	Inventory	Inventory	16
M10	HR (Staff + Payroll)	HR	17
M11	Administration	Platform	18
M12	Reports & Analytics	Reports	13
M13	Settings	Settings	13
M14	Platform Management	Platform	15

6.2 M01 — Identity & Authentication

Domain: Platform

Role Access: All users

Total Pages: 15

Page	Type	Layout	Notes
Login (Phone + OTP)	Auth	Centered card	Phone input → OTP input → verify. School logo + 'Powered by PreOne'.
Login (Email + Password)	Auth	Centered card	Email + password. Forgot password link. Remember device checkbox.
Forgot Password	Auth	Centered card	Email → reset link sent. Confirmation screen.
Reset Password	Auth	Centered card	New password + confirm. Password strength meter. Argon2 enforced server-side.
MFA Setup (TOTP)	Auth	Stepper	QR code → authenticator app → verify code → backup codes.
MFA Challenge	Auth	Centered card	6-digit TOTP input. 'Use backup code' link. 'Remember device' checkbox.
First-Time Login Onboarding	Onboarding	Stepper	Welcome → Profile setup → Security setup → Tour → Done.

Page	Type	Layout	Notes
Session Timeout Warning	Auth	Modal	Warning modal 5 min before session expiry. 'Continue session' or 'Sign out'.
Account Locked	Auth	Centered card	After 5 failed attempts. Shows unlock time. 'Contact admin' link.
User Profile	Detail	Tabs	Profile photo, name, contact, role, change password, sessions, preferences.
User Preferences	Settings	Form	Theme (light/dark/auto), sidebar mode, density, language, timezone.
Active Sessions	List	Table	List of active sessions with device, location, last active. Revoke button per session.
Change Password	Form	Card	Current password + new + confirm. Strength meter. Cannot reuse last 5 passwords.
Manage API Keys	List	Table	API keys list with name, scopes, last used, expiry. Create + revoke. 90-day rotation enforced.
Role & Permission Matrix	Matrix	Table	Roles (rows) x Permissions (columns). Check/uncheck per role. Read-only for non-super-admins.

6.3 M02 — CRM (Lead Management)

Domain: Admissions

Role Access: Sales, Admission Counselor, Branch Head, School Admin

Total Pages: 12

Page	Type	Layout	Notes
CRM Dashboard	Dashboard	Tile grid + chart	Tiles: New leads today, Lead-to-enquiry %, Hot leads, Conversion rate. Chart: leads trend.
Lead List	List	DataTable	All leads with search, filter (status, source, program, assigned to), sort, pagination, export.
Lead Detail	Detail	Tabs	Tabs: Profile, Activity Timeline, Enquiries, Documents, Tasks, Communications.
Create Lead	Form	Card form	Parent name, contact, child name & DOB, program interest, source, assigned counselor.
Edit Lead	Form	Card form	Same as Create, pre-filled.
Lead Activity Timeline	Timeline	Card timeline	All activities: calls, emails, visits, status changes. Reverse chronological.
Lead Conversion to Enquiry	Workflow	Stepper	Qualify lead → Create enquiry → Assign to admission → Notify.

Page	Type	Layout	Notes
Lead Source Report	Report	Chart + table	Leads by source (walk-in, online, referral, campaign). Conversion % per source.
Counselor Performance	Report	Chart + table	Per-counselor: leads assigned, enquiries created, admissions closed, conversion %.
Lead Assignment	Workflow	Modal	Bulk assign leads to counselor. Round-robin or manual.
Follow-up Reminders	List	DataTable	Today's follow-ups, overdue follow-ups. Snooze / complete / reschedule.
Lead Import (CSV)	Upload	Stepper	Upload CSV → Map columns → Validate → Preview → Import.

6.4 M03 — Admissions

Domain: Admissions

Role Access: Admission Counselor, Branch Head, School Admin

Total Pages: 12

Page	Type	Layout	Notes
Admissions Dashboard	Dashboard	Tile grid + pipeline	Tiles: Applications today, Pending approval, Approved this week, Rejected. Pipeline: stages.

Page	Type	Layout	Notes
Application List	List	DataTable	All applications with filter (status, program, academic year, branch), sort, pagination, export.
Application Detail	Detail	Tabs	Tabs: Applicant, Parent, Documents, Eligibility, Fee Plan, Approvals, Communications.
Create Application	Form	Stepper	Steps: Applicant info → Parent info → Documents upload → Program selection → Fee plan → Review → Submit.
Edit Application	Form	Stepper	Same as Create, pre-filled. Locked fields after approval.
Document Review	Review	Card grid	Uploaded documents in cards with preview, verify/reject buttons, rejection reason.
Eligibility Check	Workflow	Card	Age validation per program. Auto-compute from DOB. Show pass/fail with rationale.
Application Approval	Workflow	Modal	Approve / Reject / Waitlist with reason. Multi-tier approval for fee waivers > threshold.

Page	Type	Layout	Notes
Fee Plan Selection	Selection	Card grid	Available fee plans for selected program. Compare side-by-side. Select + proceed.
Admission Letter	Document	Print view	Generated admission letter with school branding. Download PDF, email to parent.
Application Status Tracking	Timeline	Card timeline	All status changes with timestamp, user, reason. Parent-visible subset via parent app.
Admission Reports	Report	Charts + table	Applications by program, conversion rate, avg time-to-approval, rejection reasons.

6.5 M04 — Student Lifecycle

Domain: Student

Role Access: Teacher, Branch Head, School Admin

Total Pages: 13

Page	Type	Layout	Notes
Students Dashboard	Dashboard	Tile grid + chart	Tiles: Total students, New this month, Active, Promoted. Chart: enrollment trend.
Student List	List	DataTable	All students with search (name, ID, parent phone), filter (class, branch, status), sort, export.

Page	Type	Layout	Notes
Student Detail	Detail	Tabs	Tabs: Profile, Parent/Guardian, Academics, Attendance, Fees, Health, Documents, Communications.
Create Student	Form	Stepper	Steps: Student info → Parent info → Medical info → Documents → Classroom → Review → Save.
Edit Student	Form	Card form	Edit student profile. Sensitive fields (DOB, parent) locked after creation — require admin.
Student Promotion	Workflow	Modal	Bulk promote at year boundary. Select class → target class → preview → confirm.
Student Transfer (Branch)	Workflow	Stepper	Initiate transfer → approve at source → approve at destination → update records.
Student Archive	Workflow	Confirm dialog	Soft-delete after 7 years per DPDP. Requires admin + reason. Reversible for 30 days.
Classroom Assignment	Workflow	Card grid	Drag-and-drop students between classrooms. Capacity warnings. Save.

Page	Type	Layout	Notes
Student Profile Photo	Upload	Modal	Upload/crop photo. Auto-resized to 256x256, 128x128, 64x64.
Medical Record	Form	Card form	Allergies, conditions, immunizations, emergency contact. Locked after doctor verification.
Student Documents	Library	Card grid	All documents (birth cert, photos, medical). Upload, preview, download, verify.
Student Reports Card	Document	Print view	Generated report card with milestones, grades, teacher remarks. Download PDF.

6.6 M05 — Academics

Domain: Academics

Role Access: Teacher, Academic Coordinator, Branch Head

Total Pages: 13

Page	Type	Layout	Notes
Academics Dashboard	Dashboard	Tile grid + chart	Tiles: Lesson plans pending, Observations today, Reports due, Milestone coverage. Chart: observations trend.

Page	Type	Layout	Notes
Curriculum Plan	Plan	Card grid	Annual curriculum per program. Themes, months, learning outcomes. Edit per month.
Lesson Plan List	List	DataTable	All lesson plans with filter (program, week, status), sort, search.
Lesson Plan Detail	Detail	Tabs	Tabs: Overview, Activities, Materials, Objectives, Assessment, AI Draft, History.
Create Lesson Plan	Form	Card form	Theme, date, duration, activities, materials, objectives. AI draft button.
Observation List	List	DataTable	All observations with filter (student, class, date, milestone), sort, search.
Create Observation	Form	Card form	Student, date, category (behavior, learning, social, motor), narrative note, photos. AI draft.
Observation Detail	Detail	Card	Full observation with photos, milestone tags, parent-share toggle.
Milestone Tracker	Matrix	Table	Students (rows) x Milestones (columns). Status: emerging / achieving / mastered. Per age band.

Page	Type	Layout	Notes
Report Card List	List	DataTable	All report cards with filter (class, term, status), sort.
Create Report Card	Form	Stepper	Select student → select term → milestones → grades → teacher remarks → review → generate. AI draft.
Report Card Preview	Document	Print view	Print-ready report card. Download PDF, email to parent, share via parent app.
AI Draft Review	Review	Side-by-side	AI-generated draft on left, editable version on right. Accept/reject/modify.

6.7 M06 — Daily Operations (Attendance + Transport)

Domain: Attendance + Transport

Role Access: Teacher, Transport Manager, Branch Head

Total Pages: 13

Page	Type	Layout	Notes
Daily Ops Dashboard	Dashboard	Tile grid + chart	Tiles: Attendance %, Late arrivals, Absentees, Transport on-time. Chart: attendance trend.
Mark Attendance (Class)	Workflow	Card grid	Student grid with present/absent/late toggles. One-tap per student. Save.

Page	Type	Layout	Notes
Mark Attendance (Transport)	Workflow	List	Trip-wise student list with pickup/drop status. GPS auto-mark.
Attendance List	List	DataTable	All attendance records with filter (date, class, student, status), sort, export.
Attendance Correction	Form	Modal	Correct attendance for a student. Reason required. Audit logged.
Absentee Report	Report	Chart + table	Absentees today, this week, this month. Trend. Frequent absentees flagged.
Transport Route List	List	DataTable	All routes with filter (branch, vehicle, driver), sort, search.
Route Detail	Detail	Tabs	Tabs: Route map, Stops, Students, Vehicle, Driver, Trip history, Live tracking.
Create Route	Form	Card form	Name, stops (address + ETA), vehicle, driver, students per stop.
Live Trip Tracking	Map	Full map	Real-time vehicle location on map. ETA per stop. Student pickup/drop status.
Trip History	List	DataTable	All trips with date, route, driver, start/end time, deviations.

Page	Type	Layout	Notes
Pickup Authorization	List	DataTable	Authorized guardians per student. OTP-based pickup verification. Override with reason + audit.
Transport Alerts	List	Card list	Speed alerts, route deviations, ETA delays. Filter by severity, date.

6.8 M07 — Parent Communication

Domain: Communication

Role Access: Teacher, Branch Head, Communication Manager

Total Pages: 12

Page	Type	Layout	Notes
Communication Dashboard	Dashboard	Tile grid + chart	Tiles: Messages today, Delivery rate, Open rate, Pending replies. Chart: message volume.
Conversation List	List	List	All conversations (1:1 with parent). Filter by unread, class, student. Search.
Conversation View	Chat	Split view	Chat interface. Messages, photos, attachments. AI reply draft button.
Broadcast List	List	DataTable	All broadcasts (1:many). Filter by channel, date, status. Search.

Page	Type	Layout	Notes
Create Broadcast	Form	Card form	Audience (class, branch, all), channels (push, SMS, WhatsApp, email), template, schedule.
Broadcast Detail	Detail	Tabs	Tabs: Content, Audience, Delivery, Opens, Clicks, Bounces.
Template List	List	DataTable	All message templates. Filter by channel, category, locale. Search.
Create Template	Form	Card form	Name, channel, category, locale, subject, body with variables. Preview.
Notification Center	List	Card list	All notifications sent. Filter by channel, status, date. Search.
Delivery Report	Report	Chart + table	Delivery rate per channel, per template, per day. Failures with reason.
Opt-out Management	List	DataTable	Parents who opted out. Channel-wise. Re-subscribe requires parent action.
AI Reply Drafts	Review	Side-by-side	AI-drafted reply on left, editable on right. Accept/edit/reject.

6.9 M08 — Finance

Domain: Finance

Role Access: Accountant, Branch Head, School Admin

Total Pages: 18

Page	Type	Layout	Notes
Finance Dashboard	Dashboard	Tile grid + chart	Tiles: Collected today, Outstanding, Overdue, DSO. Chart: collection trend.
Invoice List	List	DataTable	All invoices with filter (status, student, branch, date), sort, search, export.
Invoice Detail	Detail	Tabs	Tabs: Summary, Items, Payments, Receipt, History, Communications.
Create Invoice	Form	Stepper	Student → fee items → discounts → tax → review → generate. Auto-numbering.
Bulk Invoice Generation	Workflow	Stepper	Select class/branch → select fee plan → preview → generate all.
Payment List	List	DataTable	All payments with filter (mode, status, date, student), sort, export.
Record Payment	Form	Card form	Invoice, amount, mode (cash, card, UPI, cheque, bank), reference, date.
Payment Gateway (Online)	Payment	Payment modal	Razorpay/UPI integration. Card, UPI, net banking. Receipt auto-generated.

Page	Type	Layout	Notes
Receipt	Document	Print view	Generated receipt with school branding. Download PDF, email/SMS to parent.
Refund List	List	DataTable	All refunds with filter (status, amount, date), sort.
Process Refund	Workflow	Stepper	Invoice → refund amount → reason → approval (2-tier if > threshold) → process.
Fee Plan List	List	DataTable	All fee plans with filter (program, academic year), sort.
Create Fee Plan	Form	Card form	Program, academic year, fee items (tuition, transport, meal, activity), frequency, discounts.
Late Fee Configuration	Settings	Card form	Grace period, late fee amount/frequency, auto-apply toggle.
GST Filing Report	Report	Table + export	Monthly GST report. Tax collected, filed, pending. Export for filing.
Fee Defaulters Report	Report	Table	Students with overdue invoices. Amount, days overdue, last reminder.
Financial Statements	Report	Tabs	Tabs: P&L, Balance Sheet, Cash Flow. Per branch, per period.

Page	Type	Layout	Notes
Reconciliation	Workflow	Card grid	Match payments with bank statements. Auto-match + manual. Discrepancy flags.

6.10 M09 — Inventory

Domain: Inventory

Role Access: Store Manager, Branch Head, School Admin

Total Pages: 16

Page	Type	Layout	Notes
Inventory Dashboard	Dashboard	Tile grid + chart	Tiles: Total SKUs, Low stock, Out of stock, Stock value. Chart: consumption trend.
Item List	List	DataTable	All SKUs with filter (category, branch, stock status), sort, search.
Item Detail	Detail	Tabs	Tabs: Profile, Stock, Movements, GRN, PO, Consumption, Valuation.
Create Item	Form	Card form	Name, SKU, category, unit, reorder level, valuation method (FIFO/LIFO/avg).
Stock Adjustment	Workflow	Modal	Adjust quantity with reason (damage, loss, recount). Audit logged.

Page	Type	Layout	Notes
Purchase Order List	List	DataTable	All POs with filter (status, vendor, date), sort, search.
Create PO	Form	Stepper	Vendor → items → quantities → rates → taxes → review → approve. 2-tier if > ₹50k.
PO Approval	Workflow	Modal	Approve/reject with reason. Multi-tier for high value.
GRN List	List	DataTable	All GRNs with filter (PO, vendor, date), sort.
Create GRN	Form	Stepper	PO → received items → quantities → quality check → stock update.
Vendor List	List	DataTable	All vendors with filter (category, active), sort, search.
Create Vendor	Form	Card form	Name, GST, contact, address, payment terms, categories.
Consumption Log	List	DataTable	All consumption entries with filter (item, branch, date), sort, export.
Stock Valuation Report	Report	Table + chart	Stock value per item, category, branch. FIFO/LIFO/avg comparison.
Reorder Report	Report	Table	Items below reorder level. Suggested PO quantity. Auto-create PO button.

Page	Type	Layout	Notes
Stock Count (Audit)	Workflow	Stepper	Select location → generate count sheet → enter counts → review discrepancies → adjust.

6.11 M10 — HR (Staff + Payroll)

Domain: HR

Role Access: HR Manager, Branch Head, School Admin

Total Pages: 17

Page	Type	Layout	Notes
HR Dashboard	Dashboard	Tile grid + chart	Tiles: Total staff, Present today, On leave, Payroll pending. Chart: headcount trend.
Staff List	List	DataTable	All staff with filter (role, branch, status), sort, search, export.
Staff Detail	Detail	Tabs	Tabs: Profile, Employment, Payroll, Leave, Attendance, Documents, Performance.
Create Staff	Form	Stepper	Personal info → employment (role, designation, branch, joining) → salary structure → documents → review → save.
Edit Staff	Form	Card form	Edit staff profile. Salary changes require approval + audit.

Page	Type	Layout	Notes
Staff Onboarding	Workflow	Stepper	Create staff → setup payroll → assign role → create login → induction checklist.
Staff Offboarding	Workflow	Stepper	Resignation → notice period → handover → full & final → deactivate login → archive.
Payroll Run	Workflow	Stepper	Select month/branch → preview payslips → approve → disburse → file statutory.
Payslip List	List	DataTable	All payslips with filter (staff, month, status), sort, export.
Payslip Detail	Document	Print view	Generated payslip with school branding. Earnings, deductions, net pay. Download PDF.
Leave Request List	List	DataTable	All leave requests with filter (staff, status, type, date), sort.
Apply Leave	Form	Card form	Type, from-to, reason, coverage plan. Balance check. Approval workflow.
Leave Approval	Workflow	Modal	Approve/reject with reason. Balance impact shown. Coverage reminder.

Page	Type	Layout	Notes
Leave Balance Report	Report	Table	Per-staff leave balances. Accruals, used, available. Per leave type.
Staff Attendance	List	DataTable	Daily staff attendance with filter (date, branch, status), sort, export.
Statutory Filings	List	DataTable	PF, ESI, TDS filings with status, due date, filed date. Reminders.
Performance Review	Workflow	Stepper	Self-assessment → manager review → goals → rating → feedback → finalize.

6.12 M11 — Administration

Domain: Platform

Role Access: School Admin, Super Admin

Total Pages: 18

Page	Type	Layout	Notes
Admin Dashboard	Dashboard	Tile grid + chart	Tiles: Total tenants, Active schools, Total branches, Total users. Chart: growth.
Tenant List	List	DataTable	All tenants (schools) with filter (status, plan), sort, search. Super admin only.
Tenant Detail	Detail	Tabs	Tabs: Profile, Subscription, Branches, Users, Billing, Theme, Feature flags.

Page	Type	Layout	Notes
Create Tenant	Form	Stepper	School info → subscription plan → first branch → admin user → theme → activate.
Branch List	List	DataTable	All branches with filter (tenant, city, status), sort, search.
Create Branch	Form	Card form	Name, address, city, contact, academic year, timezone.
User List	List	DataTable	All users with filter (role, branch, status), sort, search, export.
Create User	Form	Card form	Name, email, phone, role, branch. Invite email sent.
Role List	List	DataTable	All roles with permissions count, user count. Edit permissions.
Create Role	Form	Card form + matrix	Role name, description, permission matrix (checkboxes).
Audit Log	List	DataTable	All audit entries with filter (user, action, entity, date), sort, export. Immutable.
Feature Flags	List	DataTable	All feature flags with toggle per tenant/branch. Search.

Page	Type	Layout	Notes
Integration List	List	DataTable	All integrations (payment, WhatsApp, SMS, AI) with status, last sync, config.
Integration Config	Form	Card form	API keys, webhooks, sync settings. Test connection button.
Subscription Management	Detail	Tabs	Tabs: Plan, Invoices, Usage, Add-ons, Renewal, Cancel.
Billing History	List	DataTable	All invoices with status, amount, date. Download PDF.
Design Studio	Studio	Split view	Theme customization. Live preview. Apply/Revert. (See Part 3.)
Settings (Global)	Settings	Tabs	Tabs: General, Academic Year, Workflow, Notifications, Security, Backup, API.

6.13 M12 — Reports & Analytics

Domain: Reports

Role Access: All roles (scope-based)

Total Pages: 13

Page	Type	Layout	Notes
Reports Dashboard	Dashboard	Card grid	Report categories as cards: Operations, Finance, Academics, HR, Compliance. Click to open.
Report List (by category)	List	Card grid	Available reports in category. Search, filter by frequency, format.
Report Generator	Workflow	Stepper	Select report → parameters (date range, branch, filters) → format (PDF, Excel, CSV) → schedule (one-time/recurring) → generate.
Report Viewer	Document	Print view + charts	Generated report with charts + tables. Download, email, share.
Scheduled Reports	List	DataTable	All scheduled reports with filter (frequency, status), sort. Edit, pause, delete.
Operations Report	Report	Charts + table	Attendance trends, late arrivals, absentee patterns, transport on-time.
Financial Report	Report	Charts + table	Revenue, collections, outstanding, DSO, defaulters, write-offs.
Admissions Report	Report	Charts + table	Leads, applications, conversions, time-to-approve, rejection reasons.

Page	Type	Layout	Notes
Academic Report	Report	Charts + table	Milestone coverage, observations per student, report card completion.
HR Report	Report	Charts + table	Headcount, attrition, leave utilization, payroll summary, statutory compliance.
Compliance Report	Report	Table	DPDP, POSH, NBC, RTE, FSSAI compliance status per branch.
Custom Report Builder	Builder	Split view	Drag-and-drop dimensions/measures. Filter, group, sort. Save as template.
Report Templates	List	DataTable	Saved report templates. Duplicate, edit, delete, share.

6.14 M13 — Settings

Domain: Settings

Role Access: School Admin, Branch Admin (limited)

Total Pages: 13

Page	Type	Layout	Notes
Settings Home	Navigation	Card grid	Categories: General, Academic, Workflow, Notifications, Security, Branding, Integrations, Backup, API, Audit.

Page	Type	Layout	Notes
General Settings	Settings	Card form	School name, address, contact, timezone, currency, fiscal year, locale.
Academic Year Settings	List	DataTable	All academic years with status (active/upcoming/archived). Create, activate.
Workflow Settings	Settings	Tabs	Tabs: Admissions, Finance, HR, Leave. Configure approval matrices, thresholds.
Notification Settings	Settings	Card form	Per-event channel selection. Quiet hours. Opt-out rules. Template defaults.
Security Settings	Settings	Card form	Password policy, MFA enforcement, session timeout, IP allowlist, rate limits.
Branding Settings	Studio	Split view	Redirects to Design Studio (Part 3).
Integration Settings	List	DataTable	All integrations with status, config, last sync. Edit, test, disable.
Backup Settings	Settings	Card form	Backup frequency, retention, destination (S3), encryption, restore test.
API Settings	Settings	Card form + table	API keys, rate limits, webhooks, IP allowlist. Generate, revoke, test.

Page	Type	Layout	Notes
Audit Settings	Settings	Card form	Audit retention, hash chain verification, export, alerting.
Data Export	Workflow	Stepper	Select data → format → date range → request → download link emailed. DPDP DSAR.
Data Retention	Settings	Card form	Per-entity retention periods. Auto-archive rules. DPDP compliance.

6.15 M14 — Platform Management

Domain: Platform

Role Access: Super Admin (PreOne internal)

Total Pages: 15

Page	Type	Layout	Notes
Platform Dashboard	Dashboard	Tile grid + chart	Tiles: Total tenants, MAU, API calls today, Error rate. Charts: growth, latency.
Tenant List (Platform)	List	DataTable	All tenants with filter (plan, status, region), sort, search. Suspend, terminate.
Tenant Detail (Platform)	Detail	Tabs	Tabs: Profile, Subscription, Usage, Health, Logs, Support, Impersonate.

Page	Type	Layout	Notes
Subscription Plans	List	DataTable	All plans (Starter, Growth, Enterprise, White-Label) with features, pricing, tenants on plan.
Create Plan	Form	Card form	Name, features (checkbox matrix), limits, pricing, billing cycle.
Platform Health	Dashboard	Tile grid + chart	API latency, error rate, DB connections, cache hit, queue depth, infra metrics.
Platform Logs	List	DataTable	Platform-level logs with filter (level, service, traceId), sort, search. Export.
Feature Flag Management	List	DataTable	All feature flags with toggle per tenant/branch/global. Audit log.
Webhook Management	List	DataTable	All webhook endpoints with status, last delivery, retry. Test button.
AI Gateway	Dashboard	Tile grid + chart	Tiles: Tokens consumed today, AI calls, Avg latency, Cost. Charts: usage trend.
AI Model Config	Settings	Card form	Model selection, temperature, max tokens, rate limits, cost caps.

Page	Type	Layout	Notes
Support Tickets	List	DataTable	All support tickets with filter (status, priority, tenant), sort. Assign, resolve.
Impersonate Tenant	Workflow	Confirm dialog	Super admin impersonates tenant admin for support. Audit logged. Time-limited.
Platform Audit Log	List	DataTable	All platform-level actions. Immutable. Filter, search, export.
Platform Settings	Settings	Tabs	Tabs: General, Email, SMS, WhatsApp, Payment, AI, Storage, CDN, DNS.

Part 7 — Multi-Tenant Branding & White Label

Multi-tenant branding is the architectural capability that lets 1000 schools share one database and one codebase while each having a completely different branded UI. This is not just 'logo on login page' — it is end-to-end theming: colors, fonts, layouts, dashboards, mobile app branding, and (for Enterprise) full white-label with custom app name and icon. This capability is PreOne's major USP in the Indian preschool ERP market, where competitors either (a) offer one fixed UI for all schools, or (b) require separate deployments per school. PreOne's approach — shared backend, themed frontend — gives each school a branded experience at the operational cost of a single deployment.

7.1 Concept

Example: PreOne (platform default) ↓ ABC Kids (Blue theme) ↓ Tiny Tots (Green theme) ↓ Little Angels (Orange theme). Same backend, same database, same codebase. Each school's users see only their school's branding. Cross-school data isolation enforced at the database query layer (`tenant_id` filter); cross-school branding isolation enforced at the theme layer (CSS variables from tenant's theme JSON).

Backend: Same. One PostgreSQL database, one API service, one Redis cache, one S3 bucket. `tenant_id` on every row enforces data isolation. Theme JSON per tenant enforces branding isolation.

Frontend: Different per tenant (visually). Same React codebase, same components, same routes. Only the CSS variables differ — injected from the tenant's theme JSON at app boot. The user sees a completely branded UI; the code sees `var(--primary)`, `var(--secondary)`, etc.

7.2 Branding Levels

PreOne offers four branding levels, tied to subscription plans. Each level unlocks additional customization. The highest level (White-Label) removes all PreOne branding and ships a fully client-branded mobile app.

Level	What's Included	Constraints
Standard (All plans)	Logo, school name, colors, font, welcome message on web portal	'Powered by PreOne' badge on login screen. PreOne Hub parent app shows school logo + 'Powered by PreOne'.

Level	What's Included	Constraints
Premium (Growth plan+)	All Standard + custom login background, school banner, dashboard layout customization, sidebar style	'Powered by PreOne' badge still required on web. Parent app still 'Powered by PreOne'.
Enterprise	All Premium + 'Powered by PreOne' badge removable from web portal	Parent app still 'Powered by PreOne' (requires White-Label add-on).
White-Label Add-on	All Enterprise + fully white-labeled parent app (custom app name, custom app icon, custom splash screen, no 'PreOne' mention anywhere)	Requires Enterprise plan + White-Label add-on. App store submission under client's developer account. Additional cost.

7.3 White-Label Flow

Trigger: Client purchases Enterprise + White-Label add-on.

White-Label Steps

- 1. Sales confirms white-label plan in subscription management.
- 2. Platform admin enables 'white_label' feature flag for tenant.
- 3. Tenant admin uploads: app name, app icon (1024x1024), splash screen image, custom domain (optional).
- 4. PreOne DevOps generates white-labeled mobile app build with tenant's branding (CI/CD pipeline).
- 5. App submitted to App Store / Play Store under client's developer account (client provides credentials or submits themselves).
- 6. Web portal: 'Powered by PreOne' badge removed automatically. Custom domain configured (CNAME to PreOne load balancer, TLS via Let's Encrypt).
- 7. Ongoing: app updates (new features, bug fixes) require re-submission to app stores. PreOne DevOps handles, client approves.

Technical detail: White-labeled mobile app is a separate build configuration in the React Native monorepo. The build pipeline injects tenant-specific assets (icon, splash, name) and produces a signed APK/IPA. The binary is uploaded to the client's app store account. Web portal white-label is simpler — just a feature flag that hides the 'Powered by PreOne' badge and enables custom domain routing at the CDN.

7.4 Theme Isolation

Multi-tenant branding requires multi-tenant isolation at every layer — not just the theme layer. The table below documents isolation at each layer of the stack. Cross-tenant data access is forbidden; cross-tenant branding access is impossible by construction.

Layer	Isolation Mechanism
Data	Database-level: every query filters by <code>tenant_id</code> from JWT. Cross-tenant reads require platform-admin scope and are audited. No exceptions.
Cache	Redis keys are prefixed with <code>tenant_id</code> (e.g., <code>tnt_123:session:abc</code>). Cache isolation enforced at key level. No cross-tenant cache reads possible.
Files	S3 buckets partitioned by <code>tenant_id</code> prefix (e.g., <code>s3://preone-uploads/tnt_123/students/...</code>). IAM policies enforce per-tenant access. CDN serves with signed URLs for private files.
Theme	Theme JSON per tenant. At app boot, the tenant's theme is fetched and injected as CSS variables on <code>:root</code> . Components reference <code>var(--token)</code> , so they render in the tenant's branding without any code change.
Search	Elasticsearch indices prefixed by <code>tenant_id</code> (e.g., <code>tnt_123_students</code>). Search queries filter by tenant. No cross-tenant search.

Part 8 — Implementation Roadmap

This part translates the design philosophy into an actionable engineering roadmap. It defines the design token architecture, the component library implementation plan, the Design Studio delivery sequence, and the quality gates that ensure the design system is built right. The roadmap is phased: foundations first (tokens, base components), then the Design Studio (the differentiator), then module-by-module page implementation.

8.1 Token Architecture

Design tokens are organized in four layers. Raw tokens hold primitive values. Semantic tokens alias raw tokens with meaningful names. Component tokens (optional) alias semantic tokens for component-specific tweaks. Theme overrides replace semantic token values per tenant. The cascade resolves at runtime.

Layer	Detail
Raw tokens	Primitive values: colors (hex), spacing (px), radii (px), font sizes (px), shadows (css). Stored in tokens.json (Style Dictionary format).
Semantic tokens	Aliases referencing raw tokens: --primary, --bg-card, --text-primary, --border-default. These are what components reference.
Component tokens	Optional per-component aliases: --button-primary-bg, --card-radius. Used when a component needs a slightly different value than the semantic default.
Theme overrides	Tenant theme JSON overrides semantic tokens. Platform default is the base; school/branch/user cascade on top.

Tooling: Style Dictionary transforms tokens.json into platform-specific outputs: CSS variables (web), JS constants (React), TypeScript types (engineers), Swift/Android XML (mobile), Figma tokens (designers). Single source of truth; all platforms in sync.

File structure: /packages/design-tokens/ → tokens.json (source) → build/ (generated CSS/JS/TS/Swift/XML) → consumed by /packages/ui/ (React), /apps/mobile/ (React Native), /docs/figma/ (designers).

8.2 Component Library Implementation Plan

The component library is delivered in three phases over 12 weeks. Phase 1 covers the ~40 core components needed for the first pilot module. Phase 2 adds ~25 advanced components. Phase 3 completes the library with ~21 remaining components and verifies dark mode across all.

Phase	Scope	Deliverable
Phase 1 (Weeks 1-4)	Layout components (Card, Tile, PageHeader, Sidebar, Header, BottomNav, Tabs, Modal, Drawer, Grid). Form components (TextInput, Select, DatePicker, Checkbox, Switch, Button, IconButton, FormLayout). Data components (Table, List, Badge, Avatar, EmptyState, Skeleton). Feedback (Toast, Alert, Tooltip, ConfirmDialog). Total: ~40 components.	Storybook deployed. Unit tests 90%+. Used in 1 pilot module (Identity).
Phase 2 (Weeks 5-8)	Advanced components (DataTable, Timeline, Charts, FileUpload, RichTextEditor, OTPInput, Stepper, Breadcrumb). Total: ~25 components. Refinements from Phase 1 feedback.	Storybook updated. Used in 4 pilot modules (Identity, CRM, Admissions, Student).
Phase 3 (Weeks 9-12)	Remaining components (FloatingActionButton, Sparkline, GaugeChart, Popover, LoadingOverlay, KeyVal, Divider, Spacer, Tag). Total: ~21 components. Dark mode verified across all components.	Complete library. 86 components. Used in all 14 modules.

Governance: New components require: design review (designer), architecture review (engineer), Storybook entry, unit tests, accessibility audit, documentation. Component changes require version bump (semver) and changelog. Breaking changes require migration guide.

8.3 Design Studio Delivery

Design Studio is delivered in four phases over 20 weeks. Phase 1 (MVP) ships the core customization. Phase 2 (Premium) adds advanced features. Phase 3 (Enterprise) adds badge removal and parent app branding. Phase 4 (White-Label) ships the fully white-labeled mobile app pipeline.

Phase	Scope	Timeline	Deliverable
Phase 1 (MVP)	Logo upload, color picker (primary/secondary/ accent), font selection, border radius, light/dark/auto, live preview, apply/revert.	Weeks 5-8	Functional Design Studio usable by school admins.
Phase 2 (Premium)	Login background, school banner, dashboard layout customization, sidebar style, card density, welcome message, favicon auto-gen.	Weeks 9-12	Full Design Studio per this document's spec.
Phase 3 (Enterprise)	'Powered by PreOne' badge removal, parent app branding (limited), custom domain support.	Weeks 13-16	Enterprise-grade branding.
Phase 4 (White-Label)	Fully white-labeled mobile app build pipeline. App store submission automation.	Weeks 17-20	White-Label add-on shippable.

8.4 Page Implementation Plan

Strategy: Module-by-module, in priority order. Each module: design review → component selection → page implementation → unit/integration/E2E tests → accessibility audit → design QA → release. Estimated 2-3 weeks per module with 2 engineers + 1 designer.

Module Priority

Priority	Modules	Rationale
1	M01 Identity, M08 Finance, M04 Student	Core flows — login, fees, students. Highest user impact.
2	M03 Admissions, M06 Daily Ops, M07 Communication	Daily operations — admissions, attendance, parent messaging.
3	M05 Academics, M10 HR, M09 Inventory	Operational depth — lesson plans, payroll, stock.
4	M02 CRM, M12 Reports, M13 Settings	Supporting — leads, analytics, configuration.
5	M11 Administration, M14 Platform Management	Internal — super admin and platform ops.

Quality Gates

- Design review: designer approves mockups before implementation.
- Component review: engineer verifies only library components used (no custom components).
- Accessibility audit: WCAG AA verified (color contrast, keyboard nav, screen reader).
- Responsive audit: verified at sm, md, lg, xl, 2xl breakpoints.
- Dark mode audit: all pages verified in dark mode.
- Theme audit: all pages verified with at least 3 different school themes (blue, green, orange).
- Performance audit: LCP < 2s, TTI < 3s on staging.

8.5 Accessibility

Standards: WCAG 2.1 Level AA. Verified via axe-core (automated) + manual testing (keyboard, screen reader).

Accessibility Features

- High Contrast mode: toggle increases contrast for visually impaired users.
- Large Font mode: toggle increases all text sizes by 25%.
- Reduced Motion: respects OS setting. All animations collapse to 0ms.

- Keyboard Navigation: every interactive element reachable via Tab. Focus rings visible (never removed).
- Screen Reader: all components have ARIA labels. Icons have aria-label. Images have alt text.
- Color Independence: information never conveyed by color alone (always icon + text + color).

Accessibility Testing

- axe-core in CI: blocks PR on critical/serious violations.
- Manual keyboard test: every page navigable with Tab/Shift+Tab/Enter/Space/Esc.
- Screen reader test (NVDA on Windows, VoiceOver on Mac): every page announced correctly.
- quarterly third-party accessibility audit.

Appendix A — Design Token Reference

Complete reference of all design tokens in the PreOne Fluent Metro Design System. These tokens are the single source of truth — components reference tokens, never raw values. School themes override tokens; the cascade resolves at runtime.

Color Tokens

Token	Default (Light)	Dark	Usage
--primary	#6A35FF	#9D7BFF	Primary actions, active states, brand emphasis
--primary-hover	#5A2BD9	#8566E8	Hover state for primary elements
--primary-light	#F0EBFF	#2A1F4D	Subtle backgrounds, selected item tints
--secondary	#009688	#4DB6AC	Secondary actions, supporting elements
--accent	#FF6B35	#FF8A5C	Highlights, call-to-action accents
--success	#10B981	#34D399	Success states, positive metrics
--warning	#F59E0B	#FBBF24	Warnings, pending items
--danger	#EF4444	#F87171	Errors, destructive actions
--info	#3B82F6	#60A5FA	Informational badges, tips
--bg-page	#F8F9FB	#0F1419	Page background
--bg-card	FFFFFF	#1A1F26	Card background
--bg-surface	#F1F3F5	#252B33	Secondary surfaces
--text-primary	#1A1D21	#F1F3F5	Primary text
--text-secondary	#5B6B7D	#9AA5B1	Secondary text
--text-muted	#9AA5B1	#6B7280	Muted text, placeholders

Token	Default (Light)	Dark	Usage
--border-default	#E3E7EB	#2D3440	Default borders
--border-strong	#C4CAD2	#3D4452	Stronger borders

Spacing Tokens

Token	Value	Usage
--space-0	0px	No spacing
--space-1	4px	Inline spacing, icon gaps
--space-2	8px	Tight element spacing
--space-3	12px	Input padding, list item gaps
--space-4	16px	Card padding (compact), button padding
--space-6	24px	Card padding (default), card gaps
--space-8	32px	Section padding, large card gaps
--space-12	48px	Page section gaps
--space-16	64px	Large page padding, hero spacing

Radius Tokens

Token	Value	Usage
--radius-sm	8px	Buttons, small inputs
--radius-md	12px	Inputs, dropdowns
--radius-lg	16px	Cards, modals (default)
--radius-xl	24px	Large containers, banners
--radius-full	50%	Avatars, pills

Shadow Tokens

Token	Value	Usage
--shadow-sm	0 1px 3px rgba(0,0,0,0.04)	Cards (default elevation)
--shadow-md	0 4px 12px rgba(0,0,0,0.12)	Modals, popovers, dropdowns
--shadow-lg	0 8px 24px rgba(0,0,0,0.16)	(Reserved, rarely used)

Typography Tokens

Token	Size	Weight	Line Height	Usage
--text-xs	12px	400	1.4	Captions, metadata
--text-sm	14px	400	1.4	Table cells, helper text
--text-base	16px	400	1.5	Body text, inputs
--text-lg	18px	500	1.4	Card titles, labels
--text-xl	20px	600	1.3	Page subheadings
--text-2xl	24px	600	1.25	Page titles
--text-3xl	32px	700	1.2	Tile numbers, hero
--text-4xl	40px	700	1.15	Login branding

Motion Tokens

Token	Value	Usage
--duration-fast	150ms	Hover, focus transitions
--duration-base	200ms	Modal, drawer, tile transitions
--duration-slow	250ms	Toast, page transitions
--ease-out	cubic-bezier(0.4, 0, 0.2, 1)	Standard easing

Token	Value	Usage
--ease-in	cubic-bezier(0.4, 0, 1, 1)	Exit easing
--ease-in-out	cubic-bezier(0.0, 0, 0.2, 1)	Entrance easing

Appendix B — Page Catalog Summary

Summary count of all page types across the 14 PreOne modules. Full detail in Part 6. This appendix serves as a quick-reference for scope planning and test coverage.

ID	Module Name	Domain	Page Count
M01	Identity & Authentication	Platform	15
M02	CRM (Lead Management)	Admissions	12
M03	Admissions	Admissions	12
M04	Student Lifecycle	Student	13
M05	Academics	Academics	13
M06	Daily Operations (Attendance + Transport)	Attendance + Transport	13
M07	Parent Communication	Communication	12
M08	Finance	Finance	18
M09	Inventory	Inventory	16
M10	HR (Staff + Payroll)	HR	17
M11	Administration	Platform	18
M12	Reports & Analytics	Reports	13
M13	Settings	Settings	13
M14	Platform Management	Platform	15

Total page types across all modules: 200

Appendix C — Glossary

Term	Definition
Design Token	Named value (color, spacing, radius, etc.) stored as a CSS variable. Components reference tokens, not raw values. Enables theming.
CSS Variable	Custom CSS property (--name) that holds a value. Inherited through the DOM. Foundation of PreOne's theming system.
Theme	Collection of token values that define a visual identity. Platform default, school, branch, and user themes cascade.
Design Studio	PreOne module (Settings → Design Studio) where school admins customize their theme without code.
Tile	Glanceable dashboard element showing one metric: large number + small label + view link.
Card	Universal container with title, content, and optional actions. Rounded corners, soft elevation.
White Space	Empty space around content. PreOne reserves 30%+ per screen for whitespace.
White-Label	Removal of all PreOne branding from the product. Enterprise add-on. Includes custom app name, icon, splash.
Multi-Tenant	Single deployment serving multiple customers (schools) with data and branding isolation.
Tenant	A single school. Top-level isolation boundary for data and branding.
Branch	A physical preschool location under a tenant. Can have branch-specific branding.
Bounded Context	DDD concept. A boundary within which a domain model applies. PreOne has 14 bounded contexts = 14 modules.

Term	Definition
WCAG	Web Content Accessibility Guidelines. PreOne targets Level AA.
Skeleton	Loading placeholder that mimics the shape of upcoming content. Preferred over spinners.
Toast	Transient notification at top-right. Auto-dismisses after 4 seconds.
Drawer	Side panel that slides in for filters or quick edits. Right-side by default.
Modal	Overlay dialog for focused tasks. Centered. Blocks background interaction.
Empty State	UI shown when no data exists. Illustration + headline + subtext + action button.
Design Dictionary	Single source of truth for design tokens. Style Dictionary format. Transforms to CSS/JS/TS/Swift/XML.
Style Dictionary	Open-source tool by Amazon for managing design tokens. Transforms tokens into platform-specific formats.
PWA	Progressive Web App. Installable web app with offline support. PreOne web portal is a PWA.
PreOne Hub	PreOne's parent-facing mobile app. React Native. Branded per school (limited) or fully white-labeled (Enterprise).
Sidebar	Left navigation panel. 240px expanded, 72px collapsed. Auto-hides on mobile.
BottomNav	Mobile navigation. 5 primary items + 'More'. Fixed to bottom of screen.
Breadcrumb	Navigation trail showing current location: Home / Students / Raj Patel / Edit.
Stepper	Multi-step flow indicator. Used in admissions, onboarding, payment workflows.
Live Preview	Design Studio feature showing real-time UI changes before apply.

Term	Definition
Powered by PreOne	Badge on login/bottom of branded apps. Required on all plans except Enterprise White-Label.
CSS Cascade	Order in which CSS rules apply. User preference > branch > school > platform.
Acrylic	Translucent material effect (Windows 11 Fluent). PreOne uses sparingly due to rendering cost.

Document Control & Sign-off

This UI Design Philosophy document is the authoritative reference for all visual and interaction design at PreOne. It is the foundation for the Component Library, Design Studio, and every page across all 14 modules. Compliance is mandatory for all designers and engineers. Deviations require design review approval.

Approval Matrix

Role	Name	Responsibility	Date
Head of Design	Design Council	Design philosophy and token architecture sign-off	2026-07-13
Head of Frontend Engineering	Engineering Council	Implementation feasibility sign-off	2026-07-13
Head of Product	Product Council	Product alignment and scope sign-off	2026-07-13
VP Engineering	Architecture Council	Architecture and theming architecture sign-off	2026-07-13
Head of Accessibility	Accessibility Council	WCAG AA compliance sign-off	2026-07-13
Head of Mobile	Mobile Council	Mobile (React Native) implementation sign-off	2026-07-13

Revision History

Version	Date	Author	Changes
1.0	2026-07-13	Design & Engineering Team	Initial release. Design Freeze status. PreOne Fluent Metro Design System defined. 86 components, 320 page types across 14 modules, Design Studio spec, multi-tenant branding architecture, implementation roadmap.

Document Control

This document is maintained by the PreOne Design & Engineering Team. Changes require design council approval and a new revision history entry. The document is reviewed quarterly and updated as the design system evolves. The current version is always available in the PreOne design wiki. Questions, suggestions, and deviation requests should be directed to the design council via the design Slack channel.

Distribution: All PreOne designers, frontend engineers, mobile engineers, and product managers. Classification: Internal Design Reference — not for external distribution without VP Design approval. The document is stored in the design wiki (single source of truth) and mirrored to the monorepo /docs/ folder for version control. The wiki version is authoritative when the two diverge.